EgoLife: Towards Egocentric Life Assistant

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Abstract

*We present **EgoLife**, a comprehensive project towards building an egocentric life assistant that can serve as a personal companion by understanding long-horizon, multi-person daily activities. Existing egocentric datasets and benchmarks focus on short, single-person activities, leaving a critical gap for week-long, multi-person social interactions essential for real-life assistants. To bridge this gap, we contribute three core components. **(1) The EgoLife Dataset**: a 266-hour, week-long, multimodal egocentric recording of 6 participants cohabiting in an instrumented house, captured using Meta Aria glasses alongside 15 exocentric GoPro cameras and 2 mmWave devices, enriched with dense transcripts and visual-audio captions. **(2) EgoLifeQA**: a 3,000-question multiple-choice benchmark spanning five life-assistant task categories—EntityLog, EventRecall, HabitInsight, RelationMap, and TaskMaster—with certificate-length metadata, where 2,003 out of 3,000 questions require more than 2 hours of context. **(3) EgoButler**: an agentic egocentric life assistant integrating EgoGPT, an omni-modal (video+audio) clip-level model built on LLaVA-OneVision [16] with instruction tuning on EgoIT-99K, and EgoRAG, a hierarchical retrieval-augmented generation system for ultra-long-context question answering. EgoGPT achieves state-of-the-art performance on standard egocentric benchmarks (75.4 on EgoSchema, 61.4 on EgoThink, 33.4 on EgoPlan), while EgoRAG markedly improves long-context QA, boosting accuracy on questions with certificate lengths exceeding 24 hours from 25.0 to 35.4. Our analysis highlights the critical role of personalization, multimodal caption quality, and hierarchical retrieval in advancing egocentric life assistants.*

1. Introduction

Imagine a personal AI companion that lives with you—one that remembers where you left your keys, recalls who attended last Tuesday's dinner, understands your daily habits, and can answer personalized questions grounded in your lived experience. Such an egocentric life assistant represents a transformative vision at the intersection of computer vision, natural language processing, and wearable computing. Yet, despite remarkable progress in video understanding [12, 13, 16, 32], the field still lacks the foundational data and methods to realize this vision.

Existing egocentric datasets—from EPIC-KITCHENS [9] and Ego4D [12] to EgoExo4D [13]—have driven significant advances in activity recognition, procedure understanding, and short-clip question answering. However, they share a fundamental limitation: they capture isolated, short-duration activities of single individuals, typically a few minutes per clip. Real life, in contrast, unfolds over days and weeks, involves multiple interacting people, and requires a model that can reason about evolving relationships, recurring habits, and long-horizon temporal dependencies. No existing dataset provides week-long, multi-person egocentric recordings with the annotation density needed for life-assistant research.

On the benchmarking side, QA datasets such as EgoSchema [20], EgoPlan [6], and HourVideo [5] have been instrumental for evaluating video-language models. However, these benchmarks test at most a few hours of context and do not address the ultra-long temporal reasoning—spanning multiple days—that a genuine life assistant must perform. Tasks like *What habit has this person developed over the past week?* or *How has the relationship between these two participants evolved?* demand a fundamentally different evaluation paradigm.

To address these limitations, we introduce **EgoLife**, a unified project encompassing a novel dataset, a challenging benchmark, and an agentic method for egocentric life

assistance. Our key contributions are:

EgoLife Dataset. We collect 266 hours of egocentric recordings from 6 participants who cohabit in a purpose-built "EgoHouse" for 7 days, wearing Meta Project Aria glasses [10] that capture synchronized video, audio, IMU, and gaze data. Fifteen exocentric GoPro cameras and 2 mmWave devices provide complementary viewpoints. We annotate the data with dense speech transcripts (via Whisper [26] with speaker diarization) and rich visual-audio captions generated by GPT-4o and verified by human annotators.

EgoLifeQA Benchmark. We design a multiple-choice QA benchmark with 3,000 questions spanning five task types: *EntityLog* (tracking objects and people), *EventRecall* (remembering specific events), *HabitInsight* (identifying recurring behavioral patterns), *RelationMap* (understanding interpersonal dynamics), and *TaskMaster* (procedural activity understanding). A distinctive feature is the *certificate length* metadata—the minimum look-back window required to answer each question—with 2,003 questions requiring more than 2 hours of historical context.

EgoButler System. We propose EgoButler, a two-stage agentic system. EgoGPT provides omni-modal (video+audio) clip-level understanding through instruction tuning on EgoIT-99K (99K egocentric QAs from 9 datasets) and Day-1 personalization for identity awareness. EgoRAG addresses ultra-long-context QA by constructing a hierarchical memory bank from multi-level summaries (clip/hour/day) and retrieving the most relevant evidence for answer generation.

Our comprehensive experimental analysis reveals that: (i) personalization significantly improves person-centric tasks (RelationMap: 29.6→33.6); (ii) EgoRAG dramatically boosts performance on long-context questions (>24h: 25.0→35.4; 6–24h: 26.8→38.9); and (iii) multimodal caption quality—combining both visual and audio channels—is a key driver of EgoButler performance, with human visual-audio captions achieving 45.5 average accuracy compared to 27–28 for audio-only models.

2. Related Work

2.1. Egocentric Video Datasets

Egocentric video understanding has been fueled by a series of increasingly rich datasets. Early works such as EPIC-KITCHENS [9] focused on kitchen activities with action annotations, while Ego4D [12] expanded the scope to 3,670 hours of daily-life video from 74 locations worldwide, introducing benchmarks for episodic memory, forecasting, and social interactions. EgoExo4D [13] captures skilled activities from both egocentric and exocentric viewpoints, enabling ego-exo correspondence learning. More specialized datasets include EgoExoLearn [14] for task execution, EgoPet [4] for animal-perspective understanding, and HoloAssist [33] for procedural assistance.

As summarized in Table 1, EgoLife distinguishes itself by providing *week-long* recordings (44.3 hours per participant), rich *interpersonal dynamics* among 6 cohabiting participants, *multi-modal* sensing (video, audio, IMU, gaze, 3D reconstruction), and over 400K captions. No existing dataset combines ultra-long duration with multi-person social interaction at this scale.

2.2. Egocentric Video QA Benchmarks

Video QA benchmarks have evolved from short-clip understanding towards longer temporal reasoning. EgoSchema [20] introduced the concept of "certificate length" for 3-minute clips from Ego4D. EgoPlan [6] evaluates task-completion planning, and EgoThink [7] assesses cognitive capabilities. For longer contexts, HourVideo [5] extends to 20-minute to 2-hour Ego4D clips, and EgoMemoria [36] tests episodic memory over 30-second to 1-hour segments.

Table 2 compares EgoLifeQA with these benchmarks. EgoLifeQA is unique in its ultra-long certificate lengths: 2,003 of 3,000 questions require more than 2 hours of context—a requirement unmet by any existing benchmark—enabling novel life-assistant tasks such as habit discovery and relationship analysis.

2.3. Video-Language Models

Large video-language models (VLMs) have achieved impressive performance on video understanding tasks. Commercial systems such as GPT-4o [22] and Gemini-1.5-Pro [32] demonstrate strong general-purpose video understanding. Open-source models including LLaVA-OneVision [16], LLaVA-NeXT-Video [39], LLaVA-Video [40], Qwen2-VL [8], InternVideo2 [34], and Oryx [18] offer competitive open-source alternatives. For audio-visual understanding, models like LLaMA-Omni [11] and OLA [19] integrate speech processing. EgoVLP [25] and related egocentric VLPs [2] specialize in egocentric representation learning. EgoGPT builds upon LLaVA-OneVision with an audio branch and egocentric-specific instruction tuning.

2.4. Retrieval-Augmented Generation for Video

For long-horizon video understanding, retrieval-augmented generation (RAG) has emerged as a promising paradigm. MovieChat [31] uses memory consolidation for hour-long videos, and TimeChat [27] focuses on temporal grounding. LongVA [38], LongVU [29], and LongVILA [35] extend context windows for video LLMs. EgoRAG builds on these ideas with a hierarchical multi-level memory (clip/hour/day) specifically designed for week-long egocentric recordings, enabling retrieval over hundreds of hours of footage.

Table 1. **Related Work for EgoLife Dataset — Overview of Egocentric Datasets.** For Modality, **V** denotes video, **G** denotes gaze, **IMU** denotes inertial measurement unit, **3D** denotes 3D scans. The EgoLife dataset stands out for its ultra-long egocentric footage and rich interpersonal interactions.

Benchmark	Domain	Modality	#Captions	Size (hrs)	#Clips	Dur./Clip	Multiview	Interpersonal Dynamics
EPIC-KITCHENS [9]	Kitchen	V	20K+	100	700	8.5 min	✗	✗
Ego4D [12]	Daily Activities	V, G, IMU, 3D	3.85M	3,670	9,645	22.8 min	✗	✗
EgoExo4D [13]	Skilled Activities	V, G, IMU, 3D	500K+	1,286	5,035	1 to 42 min	✓	✗
EgoExoLearn [14]	Task Execution	V, G, IMU	–	120	432	13.4 min	✓	✗
EgoPet [4]	Animal Actions	V, G, IMU	–	84	6,646	45.5 sec	✗	✗
EgoLife	Daily Life	V, G, IMU, 3D	400K+	266	6	44.3 h	✓	✓

Table 2. **Related Work for EgoLifeQA Benchmark.** The EgoLifeQA dataset is distinguished by its ultra-long video footage and certificate length, facilitating novel tasks such as habit discovery and relational interaction pattern analysis (see Figure 4 for details). **Note on Dur./Clip:** A clip is defined as a session with narrative continuity. For the EgoLife dataset, this value is derived from 266 hours of retained footage distributed across six participants.

Dataset	Source	#QAs	Size (hrs)	#Clips	Dur./Clip	Certificate Length [20]	
						Below 2h	Over 2h
EgoSchema [20]	Ego4D	5,063	250	5,063	3 min	5,063	0
EgoPlan-Bench [6]	Ego4D & EpicKitchen	4,939	–	4,939	–	4,939	0
EgoThink [7]	Ego4D	700	–	595	–	700	0
EgoMemoria [36]	Ego4D	7,026	–	629	30 s to 1 h	7,026	0
HourVideo [5]	Ego4D	12,976	381	500	20 min to 2 h	12,976	0
EgoLifeQA	**EgoLife**	3,000	266	6	44.3 h	997	2,003

3. The EgoLife Dataset & Benchmark

3.1. Data Collection

Instrumented Environment. The EgoLife dataset was collected in a purpose-built shared house ("EgoHouse") equipped with a comprehensive sensor array. Each of 6 participants wore Meta Project Aria glasses [10] capturing RGB video (1080p), binaural audio, IMU data (accelerometer/gyroscope), and eye gaze. Fifteen exocentric GoPro cameras covered common areas (kitchen, living room, study) from fixed viewpoints, and 2 mmWave radar devices on the second floor provided skeletal tracking. Figure 3 shows the 3D reconstruction of the EgoHouse with sensor placements.

Recording Protocol. Six participants cohabited in the EgoHouse for 7 days, wearing Aria glasses during waking hours ($\sim$8 hours/day per participant), yielding approximately 44.3 hours of first-person footage per participant and 266 hours total. All streams were synchronized via the EgoSync protocol to sub-second precision. Activities were entirely unscripted, encompassing cooking, socializing, studying, cleaning, and leisure activities.

Modality Synchronization and Preprocessing. Egocentric recordings were segmented into 5-minute clips for processing. EgoBlur [24] was applied to protect the privacy of non-participant individuals. Audio was denoised and synchronized with video. The primary language of interactions was Chinese, with English translations provided in all annotations for international accessibility.

3.2. Annotation Pipeline

Figure 7 illustrates our multi-stage annotation pipeline.

Speech Transcription and Diarization. We applied Whisper v3 [26] for automatic speech recognition, followed by speaker diarization [1] to assign speech to individual participants. Human annotators reviewed and corrected transcripts, yielding high-accuracy SRT files aligned to video timestamps.

Visual-Audio Caption Generation. For each 5-minute clip, we generated dense visual-audio captions by: (1) narrating clips at 0.8$\times$ speed to capture fine-grained details; (2) merging narrations with speech transcripts using GPT-4o to produce integrated visual-audio captions; (3) conducting multi-round human verification to ensure factual accuracy, temporal consistency, and completeness. This process yielded over 400K captions covering the full 266-hour dataset.

3.3. EgoLifeQA Benchmark

Question Generation. Using the dense captions and transcripts as grounding sources, we prompted GPT-4o to generate approximately 100K candidate multiple-choice questions per participant across five task categories. Human annotators then manually filtered these to 500 high-quality questions per participant (3,000 total), ensuring:

- **Answerability**: Each question has a definitive correct answer grounded in the footage.
- **Non-triviality**: Questions cannot be answered without viewing the relevant footage.
- **Distractor quality**: Wrong choices are plausible and require genuine reasoning to eliminate.
- **Metadata**: Each question is annotated with (i) an *audio-needed* flag, (ii) an *identity-needed* flag, and (iii) the *certificate length*—the minimum temporal look-back required.

Figure 1. **The Overview of EgoLife Project.** EgoLife encompasses a 266-hour multimodal dataset, the EgoLifeQA benchmark, and the EgoButler system for egocentric life assistance.

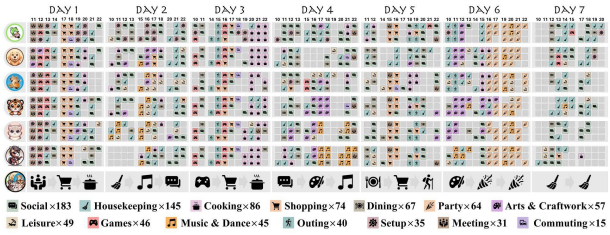

Figure 2. **The Activity Timeline of the EgoLife Dataset.** Daily activities of 6 participants over 7 days, showing rich multi-person social dynamics.

Task Categories. As illustrated in Figure 4:

- **EntityLog**: Tracking objects, tools, and people across time (e.g., *Who first used the screwdriver?*).
- **EventRecall**: Recalling specific past events (e.g., *What did Jake eat for breakfast on Day 3?*).
- **HabitInsight**: Identifying recurring behavioral patterns (e.g., *Which participant consistently wakes up the earliest?*).
- **RelationMap**: Understanding interpersonal dynamics (e.g., *Which two participants spent the most time together?*).
- **TaskMaster**: Procedural activity understanding (e.g., *In what order did Alice complete the assembly steps?*).

Certificate Length Distribution. Figure 5 and Table 2 show the certificate length distribution. Of 3,000 questions, 997 require less than 2 hours of context, while 2,003 require more than 2 hours—including many that span multiple days. This distribution validates the need for retrieval-augmented approaches over week-long memory.

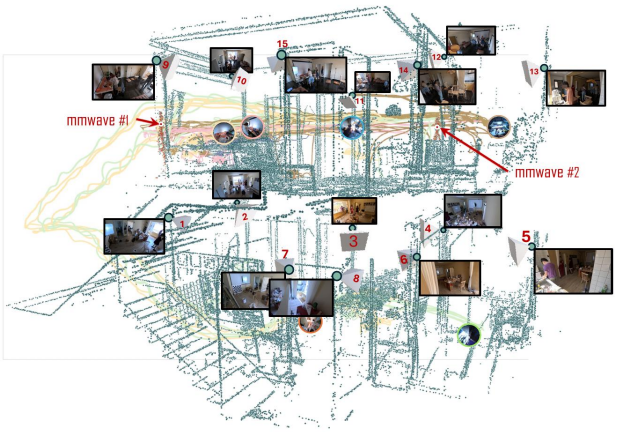

Figure 3. 3D reconstruction of the shared house using Aria Multi-MPS [10], showcasing the locations of 15 Exo cameras in the common area and 2 mmWave devices (highlighted in red) on the second floor.

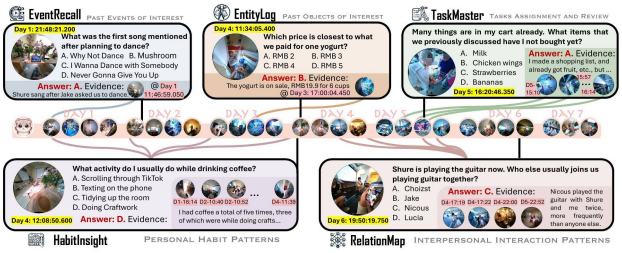

Figure 4. **Question Types and Examples in the EgoLifeQA Benchmark.** Five task categories span entity tracking, event recall, habit insight, relational understanding, and task completion.

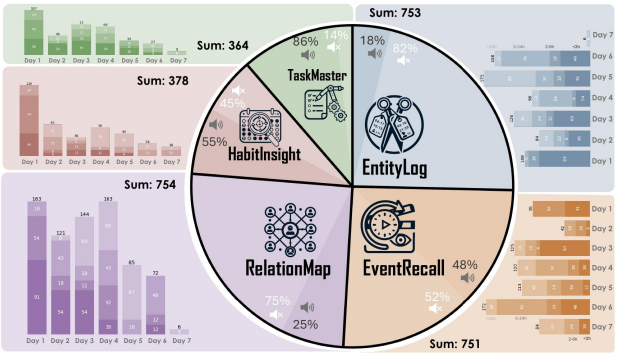

Figure 5. **Statistics of EgoLifeQA.** Distribution of certificate lengths and task categories, showing that 2,003/3,000 questions require >2h of context.

4. EgoButler: Agentic Egocentric Life Assistant

EgoButler is a two-stage agentic system for personalized, long-horizon egocentric QA. As shown in Figure 6, it integrates two components: **EgoGPT** for omni-modal clip-level understanding and **EgoRAG** for hierarchical retrieval over week-long recordings.

4.1. EgoGPT: Omni-Modal Clip Understanding

Architecture. EgoGPT extends LLaVA-OneVision [16] with an audio processing branch. Video frames are encoded by a CLIP-based visual encoder and projected into the language model embedding space. Audio is processed by Whisper v3 [26] as the speech encoder, whose outputs are projected by a lightweight MLP speech projector into the same embedding space. The language model backbone (7B parameters) receives interleaved video, audio, and text tokens to generate responses.

EgoIT-99K Instruction Tuning. To endow EgoGPT with egocentric-domain knowledge, we curate **EgoIT-99K**, a 99K-sample instruction tuning dataset derived from 9 existing egocentric datasets (Table 3). EgoIT-99K spans diverse QA types: Video Captioning (VC), Audio-Video Captioning (AVC), Multiple Choice Questions (MCQ), Multi-Round Conversations (MRC), and Image QA (IQA). Training data covers 43.16 hours across 1,529 videos, with 686 videos providing synchronized audio. This broad coverage enables EgoGPT to generalize across egocentric activities, interaction types, and environments.

Day-1 Personalization. A critical challenge for a life assistant is recognizing specific individuals across days. We address this through **Day-1 personalization**: on the first day of recording, we fine-tune EgoGPT with identity-specific data—face images, voice samples, and activity contexts for each participant. This personalization step enables EgoGPT to correctly attribute actions and speech to named individuals, which is essential for relationship-centric tasks (RelationMap) and any question requiring person identification.

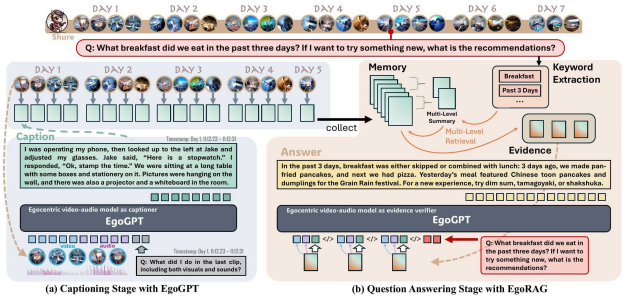

Figure 6. **The EgoButler Architecture.** EgoGPT processes individual clips with video and audio inputs, generating dense captions stored in the EgoRAG memory bank. At query time, EgoRAG retrieves relevant evidence from multi-level memory and synthesizes the final answer.

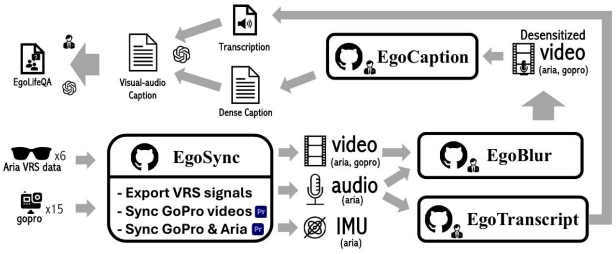

Figure 7. **The Overview of Data Process Pipeline.** From raw recordings to dense captions and QA annotations, the pipeline integrates speech transcription, visual captioning, GPT-4o merging, and human verification.

Table 3. **Dataset Composition of EgoIT-99K.** We curated 9 classic egocentric video datasets and leveraged their annotations to generate captioning and QA instruction-tuning data for fine-tuning EgoGPT, building on the LLaVA-OneVision base model [16]. **#AV** means the number of videos with audio used for training. QAs include multiple types— **VC**: Video Captioning, **AVC**: Audio-Video Captioning, **MCQ**: Multiple Choice Questions, **MRC**: Multi-Round Questions, **IQA**: Image Question-Answering.

Dataset	Duration	#Videos (#AV)	#QA	QA Type
Ego4D [12]	3.34h	523 (458)	1.41K	VC, AVC, MCQ, MRC
Charades-Ego [30]	5.04h	591 (228)	18.46K	VC, AVC, MRC
HoloAssist [33]	9.17h	121	33.96K	VC, MCQ, MRC, IQA
EGTEA Gaze+ [17]	3.01h	16	11.20K	VC, MCQ, MRC, IQA
IndustReal [28]	2.96h	44	11.58K	VC, MCQ, MRC, IQA
EgoTaskQA [15]	8.72h	172	3.59K	VC, MCQ, MRC
EgoProceL [3]	3.11h	18	5.90K	VC, MCQ, MRC, IQA
Epic-Kitchens [9]	4.15h	36	10.15K	VC, MCQ, MRC, IQA
ADL [23]	3.66h	8	3.23K	VC, MCQ, MRC, IQA
Total	**43.16h**	**1529 (686)**	**99.48K**	

4.2. EgoRAG: Hierarchical Retrieval-Augmented Generation

Given a query about week-long footage, retrieving the most relevant evidence from 44+ hours of clips per participant is a core challenge. EgoRAG addresses this with a hierarchical memory architecture and efficient retrieval.

Memory Construction. EgoRAG constructs a three-level memory bank from EgoGPT-generated captions:

- **Clip-level**: Dense visual-audio captions for each 5-minute segment, stored with timestamps and embeddings.
- **Hour-level**: Summaries of activity patterns within each hour, generated by aggregating clip captions via GPT-4o.
- **Day-level**: High-level daily summaries capturing key events, interactions, and routines across all participants.

All levels are embedded using a text encoder and stored in a ChromaDB vector database for efficient semantic retrieval.

Hierarchical Retrieval. At query time, EgoRAG performs top-k retrieval across memory levels:

1. **Coarse retrieval**: The query is matched against day- and hour-level summaries to identify relevant time windows.
2. **Fine retrieval**: Within identified windows, clip-level captions are retrieved to obtain precise temporal evidence.
3. **Answer synthesis**: Retrieved clips and captions are passed to EgoGPT (with identity context) to generate the final multiple-choice answer.

This hierarchical approach dramatically reduces the search space while preserving fine-grained temporal evidence for answering precise factual questions.

5. Experiments

5.1. Experimental Setup

Benchmarks. We evaluate EgoGPT on three established egocentric benchmarks: **EgoSchema** [20] (5,063 MCQs on 3-minute Ego4D clips), **EgoPlan** [6] (4,939 task-planning questions), and **EgoThink** [7] (700 cognitive questions). We evaluate EgoButler on **EgoLifeQA** (3,000 MCQs, 5 categories).

Baselines. We compare against commercial models (GPT-4v [21], GPT-4o [22], Gemini-1.5-Pro [32]) and open-source VLMs (LLaVA-OV [16], LLaVA-NeXT-Video [39], LLaVA-Video [40], LongVA [38], Qwen2-VL [8], InternVideo2 [34], Oryx [18], IXC-2.5 [37]).

Implementation Details. EgoGPT uses a 7B language model backbone, trained for 1 epoch on EgoIT-99K using LoRA fine-tuning on top of LLaVA-OneVision [16]. Evaluation uses 32 frames sampled at 1 FPS. Day-1 personalization uses an additional fine-tuning step on identity-specific data. For EgoRAG, we use top-k=5 retrieval at each memory level.

5.2. EgoGPT on Egocentric Benchmarks

Table 4 reports EgoGPT performance on standard egocentric benchmarks. EgoGPT (EgoIT+EgoLifeD1) achieves **75.4** on EgoSchema, **33.4** on EgoPlan, and **61.4** on EgoThink—surpassing all open-source 7B models and matching or exceeding commercial models. Even without Day-1 personalization, EgoGPT (EgoIT) scores 73.2/32.4/61.7, demonstrating the effectiveness of egocentric-specific instruction tuning. Notably, EgoGPT outperforms Gemini-1.5-Pro (72.2/31.3/62.4) on EgoSchema and EgoPlan, despite being a 7B open-source model, underscoring the value of domain-specific training data.

Table 4. **Performance of EgoGPT.** The table compares EgoGPT with state-of-the-art commercial and open-source models on existing egocentric benchmarks.

Model	#Param	#Frames	EgoSchema	EgoPlan	EgoThink
GPT-4v [21]	–	32	56.6	38.0	65.5
Gemini-1.5-Pro [32]	–	32	72.2	31.3	62.4
GPT-4o [22]	–	32	72.2	32.8	65.5
LLaVA-Next-Video [39]	7B	32	49.7	29.0	40.6
LongVA [38]	7B	32	44.1	29.9	48.3
IXC-2.5 [37]	7B	32	54.6	29.4	56.0
InternVideo2 [34]	8B	32	55.2	27.5	43.9
Qwen2-VL [8]	7B	32	66.7	34.3	59.3
Oryx [18]	7B	32	56.0	33.2	53.1
LLaVA-OV [16]	7B	32	60.1	30.7	54.2
LLaVA-Videos [40]	7B	32	57.3	33.6	56.4
EgoGPT (EgoIT)	7B	32	73.2	32.4	61.7
EgoGPT (EgoIT+EgoLifeD1)	7B	32	75.4	33.4	61.4

5.3. EgoButler on EgoLifeQA

Table 5 presents EgoButler's performance on EgoLifeQA under different model configurations. For fair comparison, we replace EgoGPT with alternative models within the same EgoButler pipeline.

Effect of Personalization. Comparing EgoGPT (EgoIT-99K) vs. EgoGPT (EgoIT-99K+D1), Day-1 personalization yields consistent improvements across all categories: average accuracy improves from 33.1 to 36.0. The largest gain is on RelationMap (29.6→33.6), which requires correctly identifying named individuals across multiple days—a task that benefits most from identity-aware representations.

Comparison with Commercial Models. EgoGPT (EgoIT-99K+D1) at 36.0 average is competitive with Gemini-1.5-Pro (36.9) despite using only 1 FPS sampling. GPT-4o without audio achieves 36.2 overall but underperforms on HabitInsight (29.5 vs. 31.1) where audio cues are valuable. LLaVA-OV without audio and personalization lags at 30.8, confirming the importance of both modalities.

5.4. Effectiveness of EgoRAG

Table 6 evaluates EgoRAG by comparing performance across certificate length buckets. We evaluate Gemini-1.5-Pro and EgoGPT on a half-hour video segment as baselines, and EgoGPT+EgoRAG over the full week-long memory.

EgoRAG provides dramatic improvements for longer certificate lengths:

- **>24h context**: EgoGPT alone scores 25.0; with EgoRAG, this improves to **35.4** (+10.4 points).
- **6–24h context**: Improves from 26.8 to **38.9** (+12.1 points).
- **2–6h context**: Improves from 29.1 to **35.7** (+6.6 points).

Interestingly, for short certificate lengths (<2h), EgoRAG shows a slight decrease (28.2→27.2), likely due to retrieval noise introducing distractors when the answer is already accessible in a short context window. These results validate the core design principle of EgoRAG: hierarchical retrieval

Table 5. **Performance comparison of EgoGPT with state-of-the-art models on EgoLifeQA benchmarks.** For a fair comparison on EgoLifeQA, EgoGPT was replaced with the corresponding models in the EgoButler pipeline to evaluate their performance under the same conditions. Models that provide captions for EgoLifeQA use 1 FPS for video sampling.

Model	#Frames	Audio	Identity	EgoLifeQA					
				EntityLog	EventRecall	HabitInsight	RelationMap	TaskMaster	Average
Gemini-1.5-Pro [32]	–	✓	✗	36.0	37.3	45.9	30.4	34.9	36.9
GPT-4o [22]	1 FPS	✗	✗	34.4	42.1	29.5	30.4	44.4	36.2
LLaVA-OV [16]	1 FPS	✗	✗	36.8	34.9	31.1	22.4	28.6	30.8
EgoGPT (EgoIT-99K)	1 FPS	✓	✗	35.2	36.5	27.9	29.6	36.5	33.1
EgoGPT (EgoIT-99K+D1)	1 FPS	✓	✓	39.2	36.5	31.1	33.6	39.7	36.0

Table 6. **Effectiveness of EgoRAG.** Integrating EgoRAG significantly enhances video-language models' performance in long-context question answering, especially for questions requiring longer certification lengths. For comparison, we evaluate Gemini-1.5-Pro and EgoGPT on a half-hour video segment, limiting their answers to this timeframe.

Model	Certificate Length			
	$< 2h$	$2h - 6h$	$6h - 24h$	$> 24h$
Gemini-1.5-Pro	27.9	14.8	25.0	18.4
EgoGPT	28.2	29.1	26.8	25.0
EgoGPT+EgoRAG	27.2	35.7	38.9	35.4

is essential for questions requiring long-horizon memory.

5.5. Ablation Study: Caption Quality

Table 7 investigates how caption quality affects EgoButler performance. We evaluate different caption sources as inputs to EgoRAG's memory bank, while keeping the answer generation model fixed (EgoGPT EgoIT-99K+D1).

Human Annotation Upper Bound. Human-generated visual-audio captions achieve **45.5** average accuracy—significantly higher than any automatic method, establishing a clear upper bound and performance gap to close.

Modality Ablation. Audio-only captions (27.2–28.1) lag substantially behind visual-only (31.2–33.6), confirming that visual information carries more discriminative content for most QA tasks. However, combining both modalities with full EgoGPT (Visual+Audio) achieves the best automatic performance (33.1/36.0), demonstrating complementary contributions.

Effect of Personalization in Captioning. Day-1 personalization (D1) consistently improves caption quality across modality configurations: Audio+Visual: 33.1→36.0; Visual Only: 31.2→33.6; Audio Only: 27.2→28.1. This confirms that identity-aware captioning translates directly into better downstream QA performance.

5.6. Qualitative Analysis

Figure 8 shows qualitative comparisons between EgoGPT and Gemini-1.5-Pro within the EgoButler framework. EgoGPT, with personalization and domain-specific training, correctly identifies participants by name and accurately recalls temporal sequences, while Gemini-1.5-Pro—without identity awareness—fails to attribute actions to the correct individuals.

Table 7. **Ablation Study on EgoGPT.** We construct different EgoRAG memory banks using generated captions from EgoGPT variants. The first three rows use captions from human annotations as a reference. All response generation models utilize EgoGPT (EgoIT-99K+D1) to ensure fair comparison. The result indicates how caption quality affects of EgoBulter performance.

Caption Source	Visual	Audio	Dataset	Avg.
Narration	✓	✓	–	31.5
Transcript	✗	✓	–	29.6
Visual-Audio Caption	✓	✓	–	45.5
EgoGPT (Audio Only)	✗	✓	EgoIT-99K	27.2
EgoGPT (Audio Only)	✗	✓	EgoIT-99K+D1	28.1
EgoGPT (Visual Only)	✓	✗	EgoIT-99K	31.2
EgoGPT (Visual Only)	✓	✗	EgoIT-99K+D1	33.6
EgoGPT (Visual+Audio)	✓	✓	EgoIT-99K	33.1
EgoGPT (Visual+Audio)	✓	✓	EgoIT-99K+D1	36.0

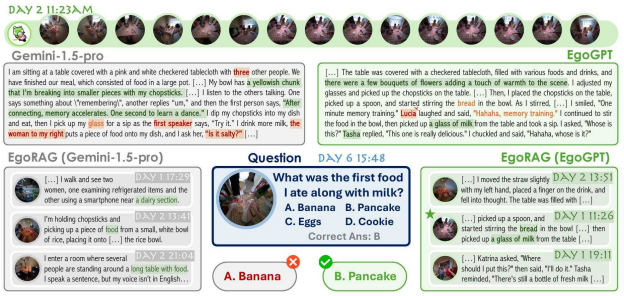

Figure 8. **Qualitative Comparison of EgoGPT and Gemini-1.5-Pro under the EgoButler Framework.** EgoGPT, with Day-1 personalization, correctly identifies named participants and recalls multi-day temporal sequences, while Gemini-1.5-Pro struggles without identity awareness.

6. Conclusion and Outlook

We have presented EgoLife, a comprehensive project advancing the vision of egocentric life assistants. Our three contributions—the EgoLife dataset (266 hours, 6 participants, 7 days), EgoLifeQA (3,000 ultra-long-context QAs), and EgoButler (EgoGPT + EgoRAG)—collectively demon-

strate that week-long, multi-person egocentric data combined with hierarchical retrieval-augmented generation can substantially advance egocentric AI.

Key takeaways include: (1) egocentric life assistance requires datasets and benchmarks that capture multi-day, multi-person dynamics—a gap EgoLife is designed to fill; (2) omni-modal (video+audio) modeling with domain-specific instruction tuning enables competitive egocentric video understanding; (3) personalization through identity-aware representations is critical for person-centric tasks; and (4) hierarchical retrieval is indispensable for ultra-long-context reasoning, delivering gains of over 10 points for questions requiring more than 24 hours of context.

Several important challenges remain open. The substantial gap between automatic (36.0) and human-captioned (45.5) performance highlights the need for better omni-modal understanding. More sophisticated reasoning over multi-day causal chains, privacy-preserving processing of sensitive personal data, and efficient online updating of the memory bank as new footage arrives are all active areas for future work. We hope EgoLife serves as a catalyst for the broader research community to build AI companions that are truly helpful in everyday life.

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